

Whole-Frame Poisson Reindexing and the Möbius-Dilation Obstruction

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Abstract

We test a proposed next transformation for the complete additive edge frame of the standard-zero-hole reduced-residue Barban–Davenport–Halberstam remainder. For a fixed unit dilation D modulo a prime q , Poisson summation has an exact reindexing $n = qr + kD$: all edge vertices share one dual integer lattice. After the divisor sum is restored, however, the dual residue packet is acted on by the multiplicative permutation $U_D : b(k) \mapsto b(kD)$. We prove an exact vector-valued whole-frame covariance identity and diagonalize all such permutations by multiplicative Fourier analysis. The resulting shared-character profile formula contains a different dual profile for every divisor. For the physical additive Fourier vector, a Gauss-sum calculation returns exactly to the nonprincipal Dirichlet-character interface of the previous scalar compiler. Finally, an exact operator-norm calculation and a $q = 5$ resonance fixture show that arbitrary profiles can align after the permutations. Thus the complete frame alone cannot collapse the divisor components into one scalar Kloosterman packet or create a power saving. This is a structural, scoped obstruction: the profile-aware prime-only theorem, the strict $1/400$ gate, and the twin-prime endpoint remain open.

1 Introduction and claim ceiling

The current prime-dynamics route studies a fixed-gap prime-pair remainder through a standard-zero-hole, prime-modulus, q -weighted reduced-residue variance. The previous stage constructed an exact complete-graph tight frame on the nonzero additive frequencies and distributed the mandatory coefficient diagonal inside the same edge cells [3]. That construction left a specific question: can one apply the Möbius and Poisson transforms to the complete oriented (d, k) frame before paying an edge or fiber triangle, and thereby expose one shared dual variable for a source-valid Kloosterman estimate?

This paper answers the algebraic part of that question and records the first failure boundary. The answer is not a simple yes or no. A shared dual integer exists for each fixed divisor component. Across divisor components, the same dual packet is seen in different multiplicative coordinates. The correct whole-frame object is consequently vector-valued. Multiplicative Fourier analysis gives a clean normal form, but that normal form is a return to the nonprincipal-character representation already identified in the earlier polarized scalar compiler [4].

The maximum justified claim is therefore

PROVED_STRUCTURAL_L1 / STOP_SCOPED_FRAME_ONLY_SAVING.

The fixed-divisor Poisson reindexing, whole-frame covariance, multiplicative profile normal form, Gauss crosswalk, and sharp alignment obstruction are proved. No prime-only BDH estimate, Kloosterman attachment, arithmetic L^2 advance, fixed-atom credit, or twin-prime theorem is claimed.

The distinction matters. A finite exact frame identity is not an arithmetic estimate, and a post-emitter Kloosterman theorem cannot be applied until the divisor profiles, the exact $(q - 2)$ diagonal subtraction, the prime shell, the four packet signs, and the physical block reassembly have all been compiled into its input type.

2 The frozen additive frame

Let $q > 2$ be prime and write

$$G_q = \mathbb{F}_q^\times, \quad N = q - 1, \quad qquad P_q = I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^*. \quad (2.1)$$

For a finitely supported sequence a , a real v , and a scale $H > 0$, define the nonzero additive-frequency vector

$$Y_q[a](k; v) = \sum_{n \in \mathbb{Z}} a_n e(vn/H) e_q(-kn), \quad k \in G_q. \quad (2.2)$$

If $e = \{k, l\} \subset G_q$, the literal edge transform is

$$T_e[a](v) = Y_q[a](k; v) - Y_q[a](l; v). \quad (2.3)$$

The complete-graph incidence vectors satisfy

$$\sum_e (e_k - e_l)(e_k - e_l)^* = NP_q. \quad (2.4)$$

Consequently the edge bilinear form is

$$\mathcal{E}_q(Y, Z) := \frac{1}{N} \sum_e (Y(k) - Y(l)) \overline{(Z(k) - Z(l))} = \langle P_q Y, P_q Z \rangle. \quad (2.5)$$

For the TPC application, (2.5) is only the internal frame layer. The physical scalar retains the outer q , the kernel localization, the prime shell, the exact coefficient-off-diagonal subtraction, and the signed packet and block reassembly. We do not replace those layers by positivity.

3 Poisson reindexing at one divisor

Let D be an integer with $(D, q) = 1$ and let F_D be a Schwartz function. Use the Fourier convention

$$\widehat{F}_D(\xi) = \int_{\mathbb{R}} F_D(x) e(-x\xi) dx, \quad e_q(x) = e(x/q). \quad (3.1)$$

The Poisson-ready component at additive vertex k is

$$Y_{q,D}(k) = \sum_{m \in \mathbb{Z}} F_D(m) e_q(-kDm). \quad (3.2)$$

Theorem 3.1 (shared dual integer at fixed dilation). *For every $k \in G_q$,*

$$\boxed{Y_{q,D}(k) = \sum_{r \in \mathbb{Z}} \hat{F}_D\left(r + \frac{kD}{q}\right) = \sum_{\substack{n \in \mathbb{Z} \\ n \equiv kD \pmod{q}}} \hat{F}_D(n/q).} \quad (3.3)$$

For fixed D , $(k, r) \mapsto n = qr + kD$ is a bijection from $G_q \times \mathbb{Z}$ to the integers not divisible by q . If

$$B_{q,D}(s) = \sum_{n \equiv s \pmod{q}} \hat{F}_D(n/q), \quad s \in G_q, \quad (3.4)$$

and $(U_D b)(k) = b(kD)$, then

$$\boxed{Y_{q,D} = U_D B_{q,D}.} \quad (3.5)$$

Proof. Poisson summation applied to $F_D(x)e(-kDx/q)$ gives the first equality in (3.3). Since D is invertible modulo q , every $n \not\equiv 0 \pmod{q}$ has the unique representation $n = qr + kD$ with $k \in G_q$. The inverse is

$$k \equiv nD^{-1} \pmod{q}, \quad r = (n - kD)/q.$$

Reindexing proves the second equality and (3.5). $\square$

Thus the fixed-divisor part of the proposed route succeeds exactly: no edge has been estimated separately, and the complete edge family uses one dual integer lattice. The price is the permutation U_D .

4 The whole-frame vector compiler

Let $\mathcal{D}$ be a finite set of unit dilations and let $c_D, d_E \in \mathbb{C}$. Set

$$Y = \sum_{D \in \mathcal{D}} c_D U_D B_D, \quad Z = \sum_{E \in \mathcal{D}} d_E U_E C_E. \quad (4.1)$$

Proposition 4.1 (whole-frame covariance). *The complete edge frame gives*

$$\boxed{\mathcal{E}_q(Y, Z) = \sum_{D, E \in \mathcal{D}} c_D \overline{d_E} \langle P_q U_D B_D, P_q U_E C_E \rangle.} \quad (4.2)$$

Moreover $U_D P_q = P_q U_D$ and the cross term is governed by the relative dilation $D^{-1}E$:

$$\langle P_q U_D B_D, P_q U_E C_E \rangle = \langle P_q B_D, U_{D^{-1}E} P_q C_E \rangle. \quad (4.3)$$

Proof. Insert (4.1) into (2.5). A multiplicative permutation preserves the constant vector and hence commutes with P_q . Also $U_D^* U_E = U_{D^{-1}E}$. This proves both identities. $\square$

Equation (4.2) is the exact whole-frame transform sought in the route. In particular, the $D \neq E$ terms are not errors. Dropping them, or replacing them by separate absolute values, would require a new estimate.

5 Shared-character profile normal form

Let $\widehat{G}_q$ denote the multiplicative character group and define the unitary multiplicative Fourier transform

$$(\mathcal{M}b)(\chi) = N^{-1/2} \sum_{s \in G_q} b(s) \overline{\chi(s)}. \quad (5.1)$$

Direct substitution shows

$$\mathcal{M}(U_D b)(\chi) = \chi(D) \mathcal{M}b(\chi). \quad (5.2)$$

The principal character coordinate is the constant direction removed by P_q .

Theorem 5.1 (shared-character profile normal form). *For Y, Z in (4.1),*

$$\boxed{\mathcal{E}_q(Y, Z) = \sum_{\chi \neq \chi_0} \left(\sum_D c_D \chi(D) (\mathcal{M}B_D)(\chi) \right) \overline{\left(\sum_E d_E \chi(E) (\mathcal{M}C_E)(\chi) \right)}}. \quad (5.3)$$

Proof. Apply the unitary transform (5.1) to (4.2), use (5.2), and delete the principal coordinate. Parseval gives (5.3). $\square$

The exact shared coordinate is now a multiplicative character χ , but the profiles $(\mathcal{M}B_D)(\chi)$ still depend on D . A scalar expression $\sum_D c_D \chi(D)$ appears only after the extra assumption that the profiles are common.

6 Gauss crosswalk to the previous interface

For a nonprincipal character χ , define

$$\tau_q(\bar{\chi}) = \sum_{x \in G_q} \bar{\chi}(x) e_q(x). \quad (6.1)$$

Proposition 6.1 (exact Gauss crosswalk). *For the physical vector (2.2),*

$$\boxed{(\mathcal{M}Y_q[a])(\chi) = \frac{\overline{\chi(-1)} \tau_q(\bar{\chi})}{\sqrt{q-1}} \sum_{q \nmid n} a_n e(vn/H) \chi(n)}. \quad (6.2)$$

Proof. Exchange the finite character sum and the sum over n . If $q \nmid n$, the substitution $x = -kn$ yields

$$\sum_{k \in G_q} \bar{\chi}(k) e_q(-kn) = \overline{\chi(-1)} \chi(n) \tau_q(\bar{\chi}).$$

If $q \mid n$, the inner sum vanishes for nonprincipal χ . Divide by $\sqrt{q-1}$. $\square$

The right side is precisely a nonprincipal Dirichlet-character transform of the original packet, up to the explicit Gauss factor and a harmless choice of whether χ or $\bar{\chi}$ names the coordinate. Hence the natural whole-frame Poisson route closes back to the V59 character representation [4]. It has not emitted the fixed-modulus Kloosterman arrays accepted by the later local engine [1].

7 Sharp alignment obstruction

The profile dependence in (5.3) cannot be removed by the graph identity.

Theorem 7.1 (sharp vector-valued alignment). *For $c = (c_D)_{D \in \mathcal{D}}$ define*

$$L_c : \bigoplus_{D \in \mathcal{D}} \mathbb{C}^{G_q} \rightarrow \mathbb{C}^{G_q}, \quad L_c((B_D)_D) = P_q \sum_D c_D U_D B_D. \quad (7.1)$$

Then

$$\|L_c\| = \left(\sum_D |c_D|^2 \right)^{1/2}. \quad (7.2)$$

Equality is attained on the centered subspace.

Proof. The adjoint is

$$L_c^* z = (\overline{c_D} U_D^* P_q z)_D.$$

Therefore

$$L_c L_c^* z = \sum_D |c_D|^2 U_D P_q U_D^* P_q z = \left(\sum_D |c_D|^2 \right) P_q z.$$

This proves the upper bound. For a unit vector $z \in \mathbf{1}^\perp$, choose

$$B_D = \overline{c_D} U_D^* z / \|c\|_2.$$

The direct-sum input has unit norm and the output has norm $\|c\|_2$. $\square$

There is a simpler coherent fixture. For unit weights, take $B_D = U_D^* z$ for one nonzero centered vector z . All output vectors then align: the sum of the individual frame energies is $|\mathcal{D}| \|z\|^2$, while the whole-frame energy is $|\mathcal{D}|^2 \|z\|^2$. Thus the ratio is exactly $|\mathcal{D}|$. This is an interface obstruction, not an asymptotic assertion about the actual Möbius coefficients.

8 Common-profile resonance and finite validation

If one imposes the additional common-profile condition $B_D = B$, then $M_c = \sum_D c_D U_D$ is diagonalized by multiplicative characters:

$$M_c \chi = \left(\sum_D c_D \chi(D) \right) \chi, \quad \|P_q M_c P_q\| = \max_{\chi \neq \chi_0} \left| \sum_D c_D \chi(D) \right|. \quad (8.1)$$

This special case still has exact resonance. For $q = 5$, $D = 2, 3$, and $c_2 = c_3 = -1$, the quadratic character satisfies $\chi(2) = \chi(3) = -1$, so its multiplier is 2, equal to the coefficient ℓ^1 mass.

The project contains an independent exact certificate for $q = 3, 5, 7, 11, 13$. It checks 1500 dual-index rows, 3016 permutation rows, all complete-graph Laplacians, and the alignment ratios. A separate Gaussian sanity script checks the continuous Poisson identity in three configurations; the maximum observed truncation error is below 10^{-12} . These checks are finite QA artifacts and do not constitute asymptotic evidence.

The exact checks can be reproduced from the repository root with

```
cd papers/tpc-209-whole-frame-poisson-mobius-obstruction
PYTHONDONTWRITEBYTECODE=1 python -B experiments/run_certificate.py --check
PYTHONDONTWRITEBYTECODE=1 python -B experiments/independent_checker.py --check
PYTHONDONTWRITEBYTECODE=1 python -O -B experiments/independent_checker.py --check
PYTHONDONTWRITEBYTECODE=1 python -B experiments/gaussian_poisson_sanity.py
```

Table 1: Finite certificate summary.

quantity	value	status
prime moduli	3, 5, 7, 11, 13	exact
dual reindex rows	1500	exact
permutation-matrix rows	3016	exact
$q = 5$ coherent energy ratio	2	exact
Gaussian Poisson cases	3	numerical sanity
maximum Gaussian error	4.5×10^{-16}	numerical sanity

9 Source boundary and route decision

The complete edge frame from TPC-208 is an exact finite-dimensional identity. The Poisson step in Theorem 3.1, the covariance in Proposition 4.1, the spectral form in Theorem 5.1, and the Gauss crosswalk in Proposition 6.1 are proved directly here. No external theorem is used to promote them to an asymptotic estimate.

The current source lock contains a general-sequence BDH architecture and post-emitter Kloosterman estimates [2, 1]. The latter accept fixed-modulus bilinear arrays after a valid emitter and norm ledger have been supplied. Equations (5.3) and (6.2) are upstream of that input: they retain divisor-dependent profiles, the exact character family, and the physical diagonal boundary. No locked source proves the required prime-only, signed, block-reassembled profile bound.

The first fatal for this particular route is therefore

NO_FRAME_ONLY_SCALAR_EMITTER: Poisson gives one shared dual lattice per dilation, but dilation permutations survive; character diagonalization returns to V59, and arbitrary profiles align sharply.

This is a scoped stop, not a global nonexistence theorem. The smallest repaired target is a profile-aware estimate for (5.3) on the actual Möbius/hybrid packets, preserving the $(q - 2)$ subtraction, prime shell, kernel localization, packet signs, and physical normalization. That target remains open and is the natural next route question.

10 Conclusion

The complete TPC-208 edge frame does support a legitimate whole-frame Poisson reindexing, but only in a vector-valued sense. The exact shared integer $n = qr + kD$ survives all edges for one divisor. Across divisors, the residual permutations are not cosmetic: they are the representation-theoretic content of the frame. Multiplicative Fourier exposes a shared character coordinate and simultaneously shows the exact return to the earlier V59 character packets. The sharp alignment theorem rules out obtaining a scalar emitter or a power saving from frame algebra alone.

The result advances the route by replacing a vague “look for a shared dual variable” instruction with a precise profile-aware interface and a precise obstruction. It leaves the arithmetic Gate B status unchanged:

FULL_GATE_B=OPEN, STRICT_1/400=UNPAID, ARITHMETIC_ADVANCE=NO.

References

- [1] Valentin Blomer and Alexandru Pascadi. Bilinear forms with Kloosterman sums via quadratic characters, 2026. Version 1.
- [2] Adam J. Harper. Simple Barban–Davenport–Halberstam type asymptotics for general sequences, 2024. Version 1.
- [3] Liang Wang. A canonical additive edge frame for zero-hole reduced-residue BDH remainders. TPC-208, internal prime-dynamics-theory paper, 2026. Structural L1 release.
- [4] Liang Wang. Polarized local BDH scalar compiler. Internal prime-dynamics-theory repository artifact, 2026. V59, research/tpc-big-road.